An End-to-End Computer Vision Pipeline for Lower-Limb Wound Classification, Detection, and Segmentation

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Abstract

Lower-limb and foot wounds need consistent visual assessment for clinical screening, localization, and treatment monitoring. Manual review is time-consuming and observer-dependent, particularly when wound boundaries are irregular or image quality is poor. This paper presents an end-to-end medical computer vision pipeline for automated wound assessment using the Lower Limb and Feet Wound Image Dataset. The system is built across four project stages: dataset preparation and annotation tooling, binary wound screening, wound-region object detection, and pixel-level wound segmentation. A MobileNetV3-Large transfer-learning classifier was trained to distinguish normal and wound images, reaching 87.33% validation accuracy and 88.61% wound-class F1-score. A YOLOv8n detector localized wound regions at a test mAP50 of 49.27% with precision of 52.92%. Finally, a U-Net segmentation model with a ResNet-34 encoder produced binary wound masks with 50.31% IoU, 66.94% Dice coefficient, and 92.68% pixel accuracy on the held-out test split. Classification provides reasonable screening performance, while detection and segmentation produce useful spatial outputs that remain sensitive to wound shape variability, annotation quality, and dataset imbalance.

Keywords: medical computer vision, wound assessment, image classification, object detection, semantic segmentation, YOLOv8, U-Net

1. Introduction

Chronic and acute wounds of the lower limbs place a significant burden on clinical workflows because they require repeated inspection, measurement, and documentation over time. Visual wound assessment is the standard approach for estimating tissue condition, wound extent, and healing progress. Manual assessment, however, can be slow and inconsistent, particularly when dealing with irregular wound borders, mixed tissue appearance, and images captured under different lighting conditions or with consumer-grade cameras.

Computer vision can meaningfully support wound-care workflows by automating three related tasks. Image classification answers whether a given image contains a wound at all. Object detection goes further and localizes the wound region using bounding boxes. Segmentation provides the finest level of detail by estimating the wound boundary at the pixel level, which is directly useful for area measurement and longitudinal monitoring. This project implements all three tasks and integrates them into a local web dashboard for reproducible experimentation.

The paper documents the full four-week pipeline: dataset selection and annotation (Week 1), binary classification (Week 2), object detection (Week 3), and pixel-level segmentation (Week 4). The final system uses transfer learning for classification, YOLOv8 for detection, and U-Net with a ResNet-34 encoder for segmentation.

2. Related Work

Deep convolutional neural networks have become the default choice for medical image classification and dense prediction tasks. ResNet introduced residual connections that allow much deeper networks to train without degradation and remains a widely used backbone for transfer learning [?]. MobileNetV3 brought efficient image classification through neural architecture search and lightweight convolutional design, making it practical for settings with limited compute resources [?].

For object detection, the YOLO family is popular because it handles both localization and classification in a single forward pass. YOLOv8, maintained by Ultralytics, includes compact variants such as YOLOv8n that work well for local GPU and CPU experimentation [?]. For medical segmentation tasks, U-Net remains a strong and well-understood baseline because its encoder-decoder structure combines high-level semantic context with fine-grained spatial detail through skip connections [?]. The segmentation component of this project follows the broader transfer-learning strategy by pairing U-Net with a ResNet encoder.

The dataset used throughout this work is the Lower Limb and Feet Wound Image Dataset, published as an open-access *Data in Brief* resource that includes wound images, healthy feet images, and corresponding segmentation masks [?]. Its combination of classification and segmentation labels makes it well suited for a multi-task wound assessment pipeline.

3. Dataset and Annotation

3.1 Dataset Source

The project uses the Lower Limb and Feet Wound Image Dataset. Based on the project documentation, the dataset contains 2,686 wound images with segmentation masks alongside 2,757 healthy feet images, totaling 5,443 JPG images at 331 × 331 pixels. Wound images span eight clinical categories: diabetic, pressure, trauma, venous, surgical, arterial, cellulitis, and miscellaneous.

3.2 Annotation Tool

Week 1 involved building a custom Tkinter-based annotation tool rather than relying on an external labeling platform. The tool supports bounding-box drawing, wound action-region labeling, wound type selection, severity scoring on a 0 to 7 scale, clinical note fields, and export to CSV and JSON/COCO-style formats. This tool established the annotation workflow used to support detection and reporting in later stages.

3.3 Task-Specific Dataset Preparation

For the Week 2 classification task, the prepared dataset contained 1,500 samples: 1,000 wound images and 500 normal images. A stratified 80/20 train-validation split was applied.

For the Week 3 detection task, annotations were converted to YOLO format using normalized bounding boxes. The prepared detection dataset contained 2,673 wound samples divided into 1,871 training images, 534 validation images, and 268 test images.

For the Week 4 segmentation task, 1,000 wound images from `data/wound_main/` were paired with 1,000 binary masks from `data/wound_mask/`. The data was split into 70% training, 15% validation, and 15% testing. The segmentation manifest was stored at `data/cleaned_week4/segmentation_manifest.csv`.

4. Methodology

4.1 Week 2: Classification

The classification module predicts two classes: normal and wound. The primary model is MobileNetV3-Large with transfer learning, where the feature extractor is kept frozen during training to improve stability and reduce compute cost. Input images are resized to 331 × 331 pixels. Training uses AdamW [?] with a learning rate of 3×10^{-3}, weight decay of 10^{-4}, cosine annealing, and standard cross-entropy loss. Data augmentation includes horizontal flips, vertical flips, random rotation, color jitter, and ImageNet normalization. The class imbalance between wound (1,000) and normal (500) samples was handled through weighted random sampling during training.

4.2 Week 3: Object Detection

The detection module uses YOLOv8n initialized from the `yolov8n.pt` checkpoint pretrained on COCO. The model was configured with 512-pixel input images, batch size 8, automatic mixed precision, and early stopping patience of 15 epochs. The Week 3 scripts cover dataset preparation, model training, evaluation on validation and test splits, prediction image generation, and export to the React dashboard. The live detection API runs at `http://127.0.0.1:5001` and serves wound bounding-box predictions through the `/week3/detect` endpoint.

4.3 Week 4: Segmentation

The segmentation module uses U-Net with a ResNet-34 encoder. The model outputs a single-channel binary mask where wound pixels are labeled 1 and background pixels are labeled 0. Input images are resized to 256×256 pixels. Training uses AdamW with learning rate 10^{-3}, weight decay 10^{-4}, cosine annealing, and a combined loss consisting of 50% binary cross-entropy with logits and 50% Dice loss. Augmentation includes horizontal flips, vertical flips, 90° rotations, color jitter, and ImageNet normalization. The best checkpoint is saved to `week4/weights/best_seg_model.pth`. The segmentation API runs at `http://127.0.0.1:5002` and exposes `/week4/segment`, `/week4/results`, and `/week4/health`.

5. Experiments and Results

5.1 Classification Results

The Week 2 classifier was evaluated on 300 validation samples. Table **??** summarizes the main metrics. The model reached 87.33% accuracy, with wound-class recall of 89.50% and F1-score of 88.61%. Macro F1 was 86.94%, which is slightly lower because the normal class is harder to classify given its smaller share of the training data. Overall, transfer learning with MobileNetV3-Large gives reasonable binary screening performance, though the class imbalance introduces some confusion between peri-wound and normal tissue.

Table 1: Week 2 Binary Classification Results (300 validation samples)

Metric	Value	Notes
Accuracy	87.33%	300 validation samples
Precision	87.74%	Wound-class precision
Recall	89.50%	Wound-class recall
F1-score	88.61%	Wound-class F1
Macro F1	86.94%	Normal and wound average

5.2 Detection Results

The Week 3 YOLOv8n detector was evaluated on both validation and test splits. Table **??** reports precision, recall, mAP50, and mAP50–95. Test mAP50 reached 49.27%, while mAP50–95 was 22.11%. These numbers indicate that the detector learned meaningful wound localization but still struggles with tighter IoU thresholds and difficult wound appearances.

Table 2: Week 3 YOLOv8n Detection Results

Split	Precision	Recall	mAP50	mAP50–95
Validation	52.74%	48.50%	48.31%	22.17%
Test	52.92%	47.76%	49.27%	22.11%

5.3 Segmentation Results

The Week 4 model was evaluated on 401 held-out test samples using a sigmoid threshold of 0.5. Table **??** lists the pixel-level metrics. The model achieved 50.31% IoU and 66.94% Dice coefficient, with pixel accuracy of 92.68%. Pixel accuracy is high primarily because background pixels dominate each image, so IoU and Dice are the more informative indicators of actual wound-mask quality.

Table 3: Week 4 U-Net Segmentation Results (401 test samples, threshold = 0.50)

Metric	Value	Notes
IoU	50.31%	Primary spatial overlap metric
Dice coefficient	66.94%	Harmonic overlap measure
Pixel accuracy	92.68%	Background-dominated
Precision	66.26%	Wound-pixel precision
Recall	67.63%	Wound-pixel recall

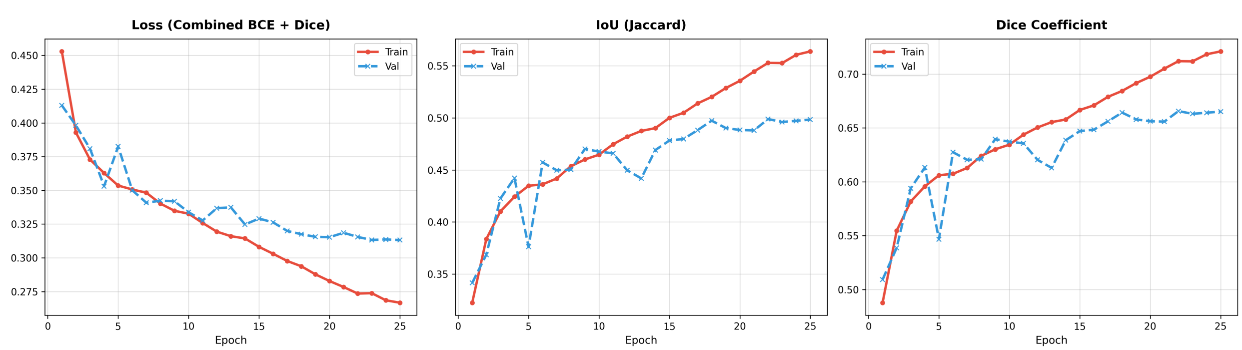


Figure 1: Week 4 segmentation training curves showing loss, IoU, and Dice trends across epochs.

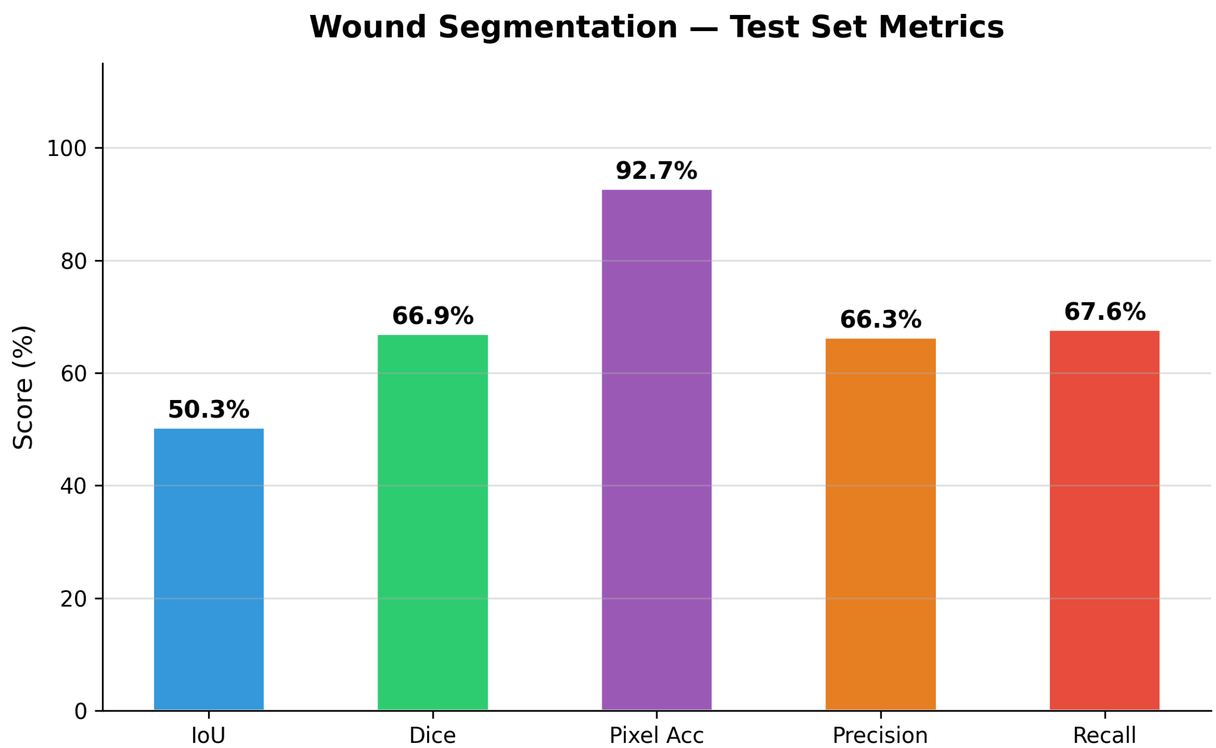


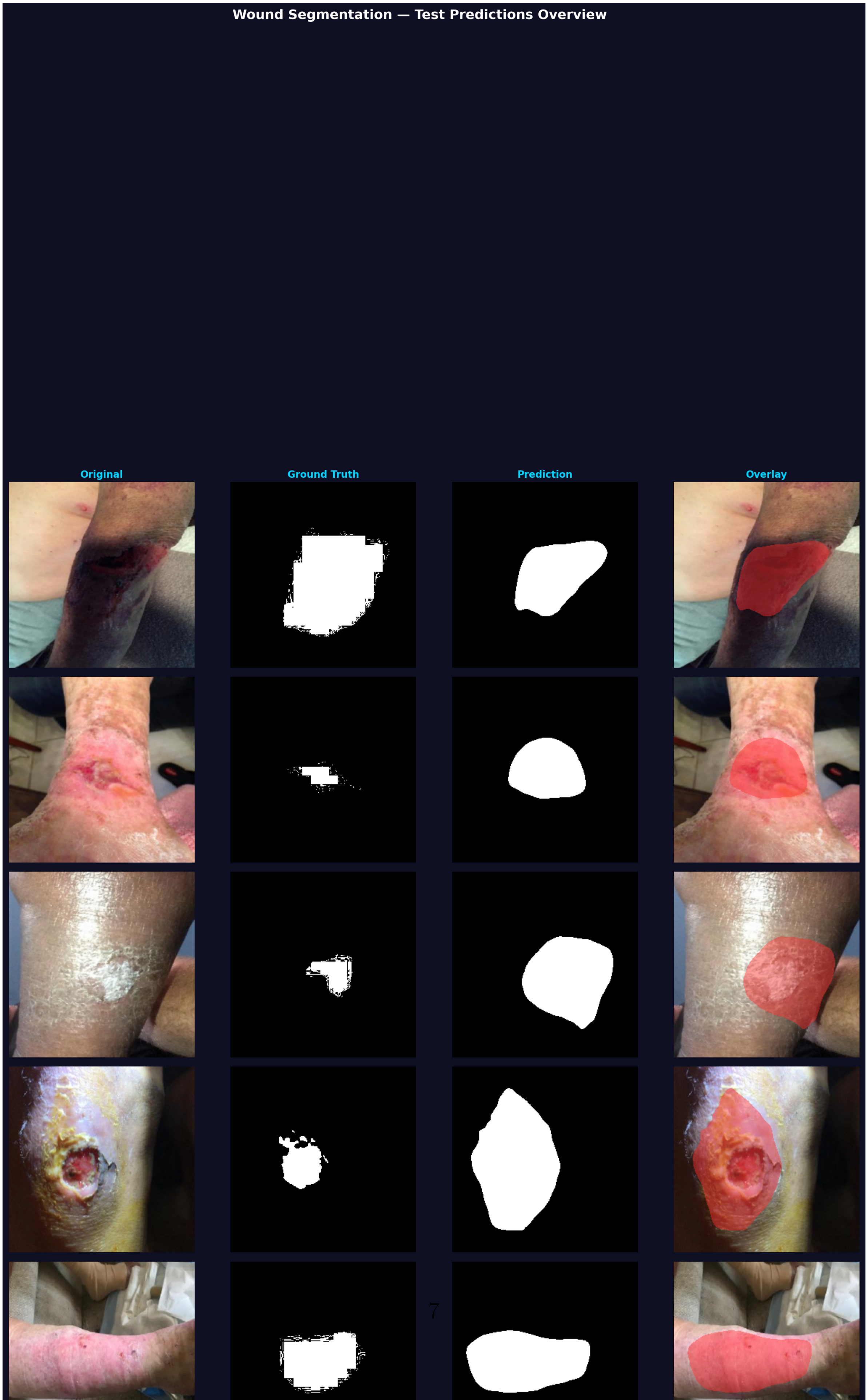
Figure 2: Week 4 test-set segmentation metrics visualized as a bar chart.

6. Discussion

The pipeline shows that binary wound screening is a tractable problem with transfer learning, though the results are more modest than near-perfect numbers sometimes reported on cleaner benchmark datasets. The 87.33% accuracy and 88.61% wound F1-score are in line with the difficulty introduced by a 2:1 wound-to-normal class ratio, variable image quality, and the visual overlap between peri-wound tissue and normal skin. Wound recall (89.50%) is higher than precision, which is the preferred trade-off for a screening tool where missing a wound is more costly than a false alarm.

The detection results are moderate and broadly consistent with the segmentation performance. Wound localization is inherently harder than image-level classification because the bounding box must capture wound extent despite irregular shapes, variable scale, and ambiguous boundaries between wound bed, peri-wound tissue, and surrounding skin. The gap between mAP50 (49.27%) and mAP50–95 (22.11%) suggests the detector can find the approximate wound region but struggles to tightly match the reference annotations.

Segmentation provides the most detailed spatial output. A Dice score of 66.94% represents useful mask overlap for a single-model baseline, while the IoU of 50.31% leaves clear room for improvement. Pixel accuracy of 92.68% is less meaningful here because background pixels


Wound Segmentation — Test Predictions Overview
Original
Ground Truth
Prediction
Overlay

make up the majority of each image. Qualitatively, the model can identify broad wound regions but may under-segment small wounds, smooth over fine boundaries, or confuse red peri-wound tissue with the wound bed itself. The fact that all three tasks perform at similar levels points to shared dataset constraints as the underlying bottleneck rather than individual model choices.

There are several clear limitations. The project uses local splits without external validation, so performance on images from different hospitals, cameras, or patient populations is unknown. Annotation quality directly bounds detection and segmentation performance, and any inconsistency in the ground-truth masks propagates into the training signal. The segmentation model also only produces binary masks; separating wound bed, necrotic tissue, and peri-wound regions would be considerably more useful clinically. Finally, the current system is a research prototype and should not be applied in any clinical setting without rigorous prospective validation.

7. Conclusion

This paper described an end-to-end wound computer vision pipeline covering annotation, binary classification, object detection, and pixel-level segmentation. The MobileNetV3-Large classifier from Week 2 reached 87.33% accuracy and 88.61% wound F1-score, providing practical screening performance under real-world class imbalance conditions. The YOLOv8n detector from Week 3 localized wound regions at 49.27% test mAP50. The U-Net model from Week 4 produced wound masks at 50.31% IoU, 66.94% Dice, and 92.68% pixel accuracy. The three performance levels align with one another and reflect shared dataset challenges rather than isolated failures in any single module. Together, these components form a functional workflow for automated wound image analysis and a starting point for future work on clinical measurement, severity scoring, and longitudinal monitoring.

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